Data Driven App:

Development Process

Abstract

Link to the github repository: https://github.com/CodeLab-II-22-23/data-driven-app-SolomonMoulangLewis

For this assignment we have been asked to create an interactive GUI application that uses data gathered from an API. We have been asked to create the application with C++, using the openFrameworks library and an API. For my project I decided to use the MealDB API, making use of the ofxJSON addon in openFrameworks. The idea behind my application is to allow the user to enter the ingredients in their cupboard. Upon entering the ingredients and pressing search, the user will be presented with a list of meals. If they click a meal they are then presented with the ingredients they have, the ingredients they need and the recipe for the meal.

Project Plan

To aid time management for this project I created a Gantt chart [1]. The Gantt chart helped me manage my time for the project by providing a clear and visual representation of the tasks and deadlines involved. It allowed me to break down the project into smaller, manageable chunks and to see the dependencies between tasks. I helped me to set clear deadlines for each task and track my progress throughout the project. The Gantt chart also helped me identify potential bottlenecks and allocate resources effectively.

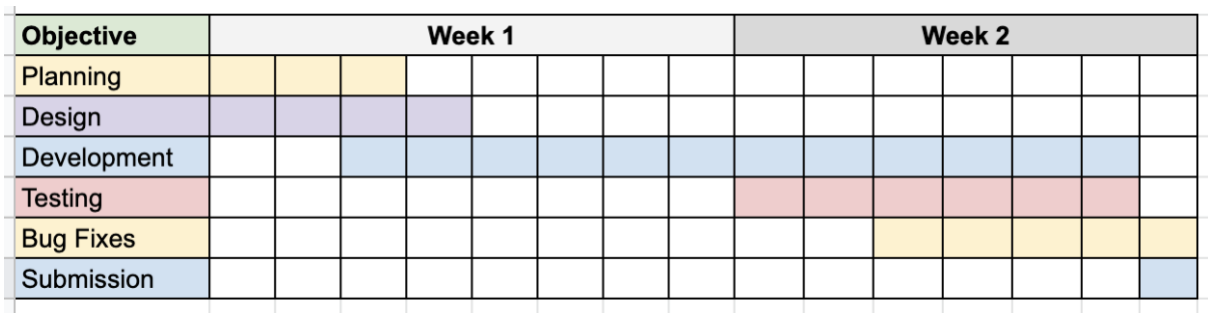

Figure 1. Gantt chart

Evidence of Design

Concepting the Backend

I created a flowchart [2] that helped me understand how data moves through the application by providing a visual representation of the different steps and processes involved. It helped me to see the flow of data from input to output, as well as the different decision points and branches that the data may take. Additionally, the use of symbols and labels in the flowchart made it easy to understand the purpose and function of each step in the process. This helped me to understand the data flow of the application and identify areas for improvement.

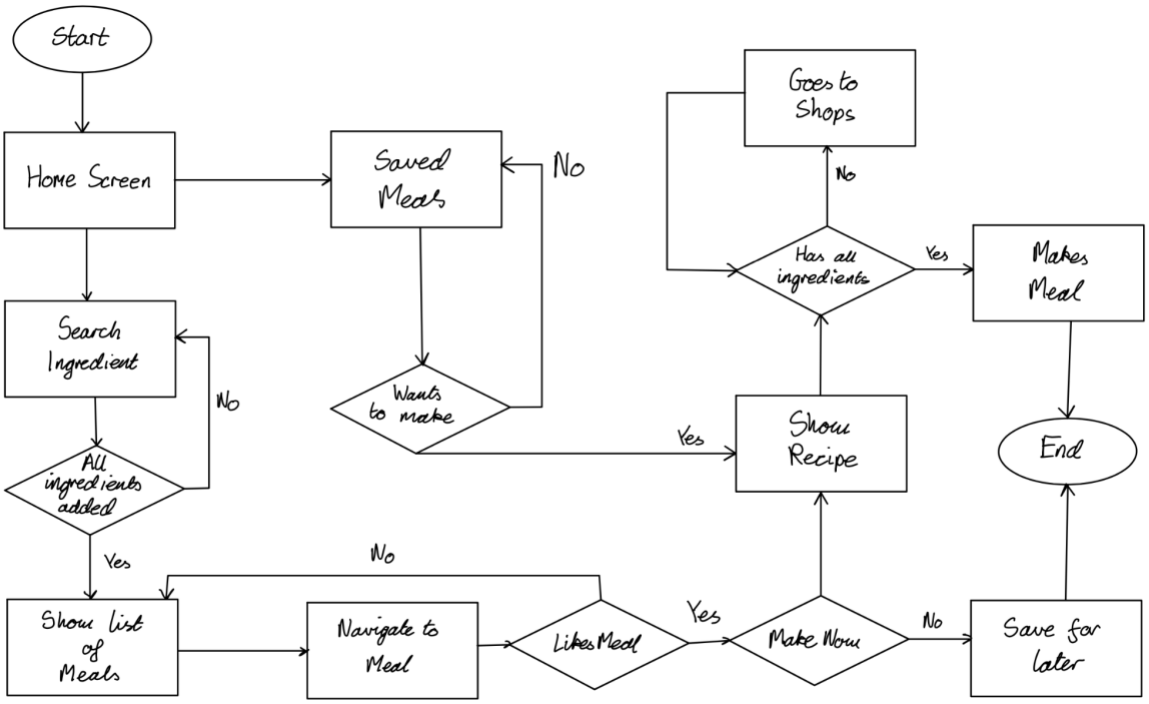

Figure 2. Flowchart

Prototyping Wireframe

I initially created low fidelity prototype wireframes of the different screens of the application [3-5]. These sketches were used to test and explore different design concepts, such as the placement of buttons and the overall layout of the application. My goal was to quickly iterate and test different ideas, rather than to create a polished final product.
I then took the low-fidelity prototype I had previously made and added more details to create a design that I liked for the front-end [6-8].

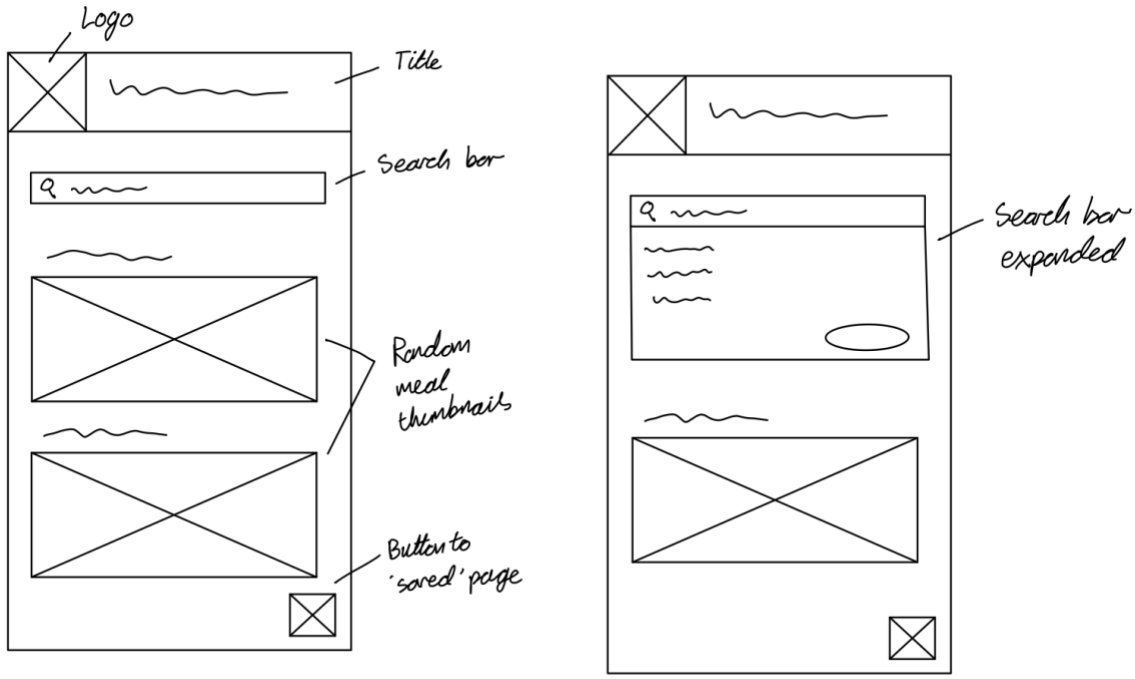

Figure 3. Low Fidelity Prototype Wireframe - Home Screen

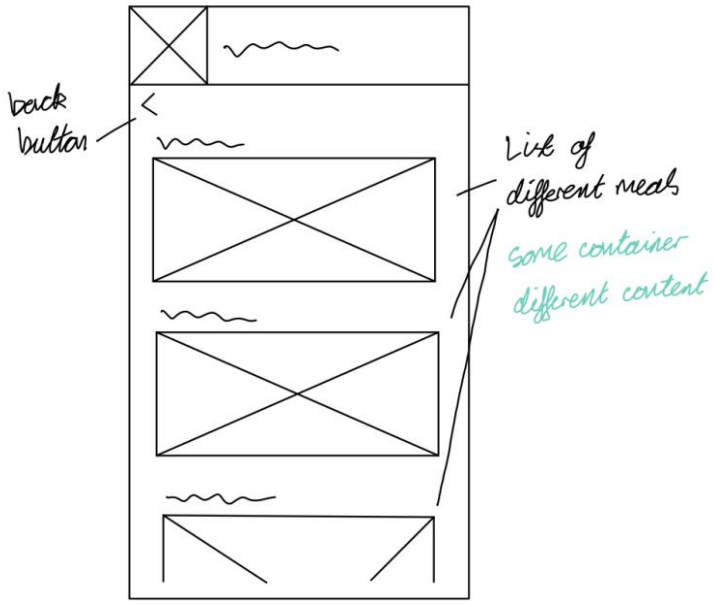

Figure 4. Low Fidelity Prototype Wireframe - Search Screen

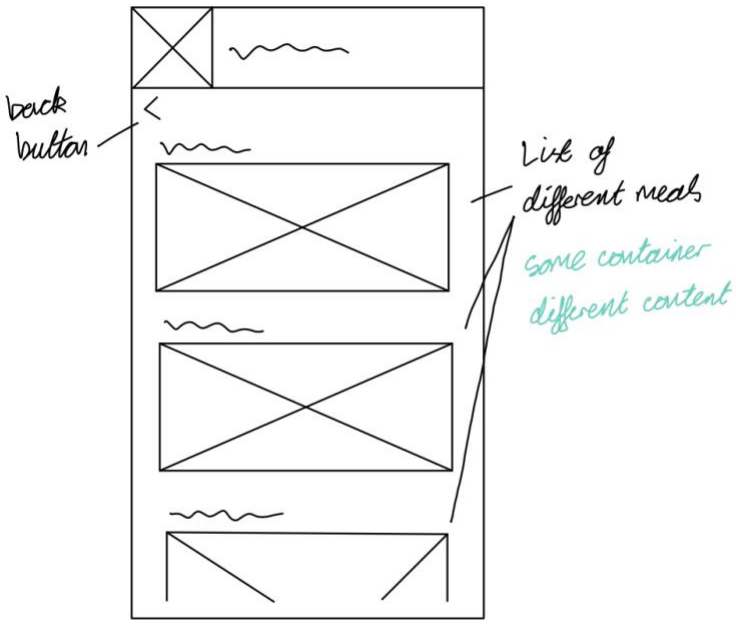

Figure 5. Low Fidelity Prototype Wireframe - Recipe Screen

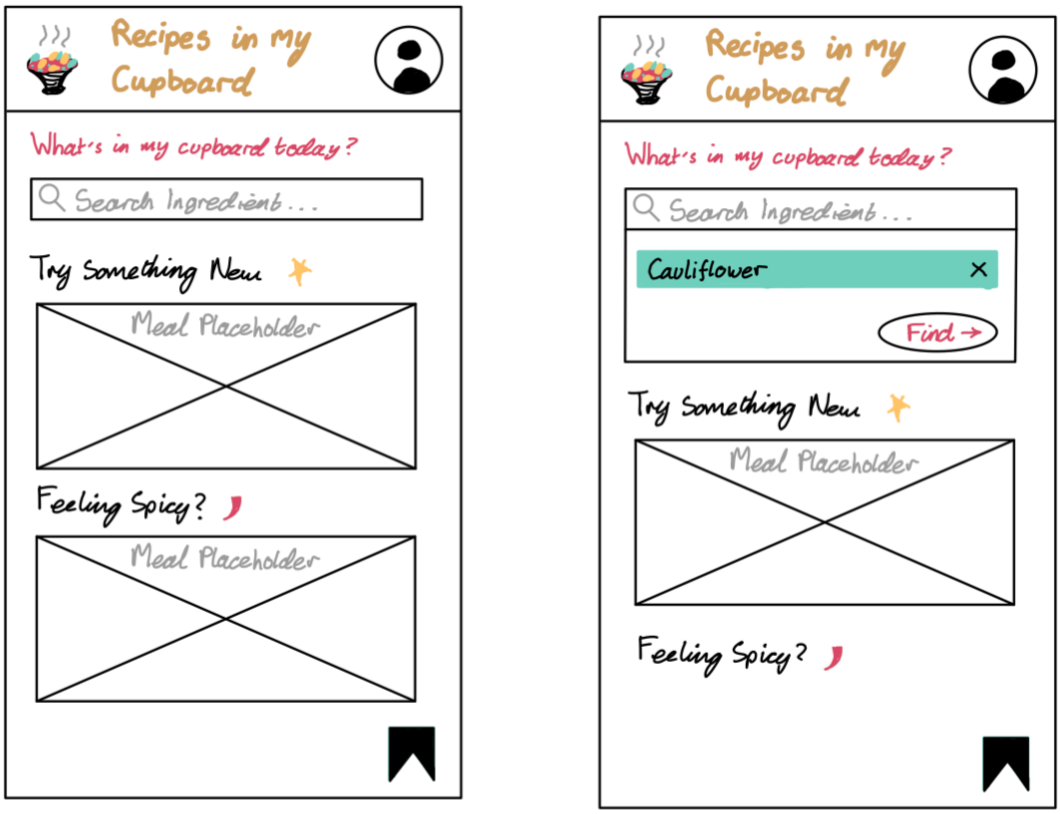

Figure 6. Added details to Low Fidelity Prototype Wireframe - Home Screen

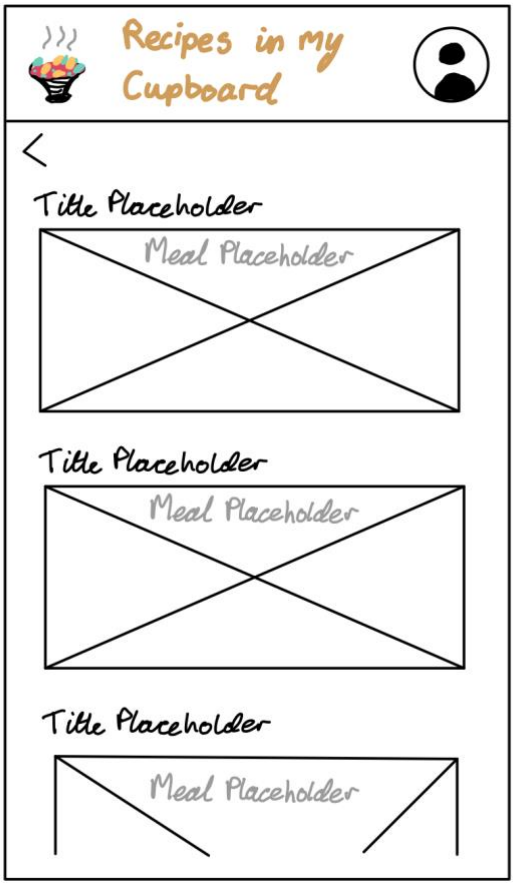

Figure 7. Added details to Low Fidelity Prototype Wireframe - Search Screen

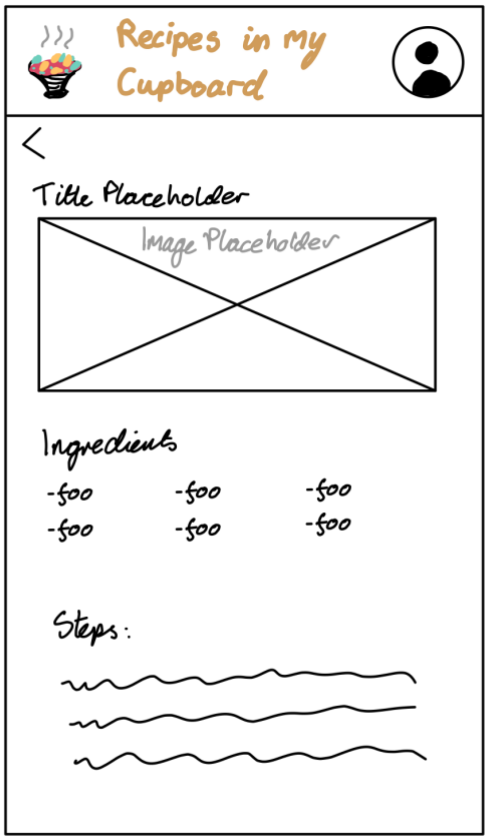

Figure 8. Added details to Low Fidelity Prototype Wireframe - Recipe Screen

Technical Description

Concepting Application Structure

I designed a flowchart [9] using an object-oriented approach to help me understand the different objects and how they interact in my project. I started by identifying the key objects in my project and their properties and methods. I separated these into three different categories; screens, components and api. I then created a visual representation of these objects and how they interact with one another. This approach helped me to identify any potential issues or areas of improvement in the design and make the necessary adjustments. Using a flowchart helped me to better understand the structure and functionality of my application.

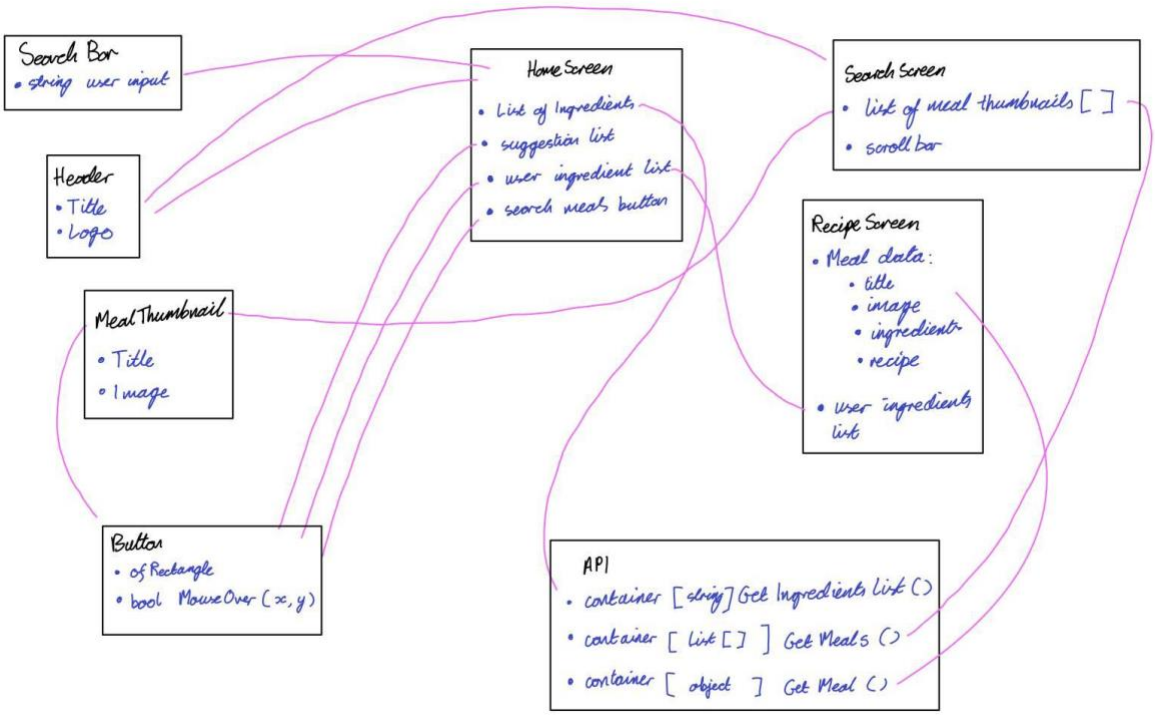

Figure 9. Application Structure Concept

Technical Walkthrough

Link to the technical walkthrough video:

https://drive.google.com/file/d/1rBnbxP0Wfnu0TG0mHsVX2qmKD3XeFjPr/view?usp=share_link

Testing

State	Scenario	Expected Result	Actual Result	Pass/ Fail
Home screen	Entering text	Text should appear in the search bar	As expected.	Pass
Home screen	Entering text	Each key pressed should correspond with the key shown	Special character keys showing unexpected results. E.g. command key = f	Fail
Home screen	Entering text	Suggestion list should show when text is entered on the keyboard A maximum of five results should show beneath the search bar. Each result should be unique and correspond with the input text	As expected	Pass
Home screen	Click ingredients suggestion	Click on any of the ingredients suggestions The result should: 1. Delete the contents of the search bar 2. Hide the suggestions list 3. Show the users list of ingredients	As expected	Pass
Home screen	Deleting ingredient from the users list	Click on any of the ingredients in their list, removing the contents	As expected	Pass
Home screen	Clicking search meals button	Should cause the application to navigate to the search screen	Passed. On the first try the meal database and return 'null' from the search.	Pass
Search screen	Content height is more than the container height	Scrollbar should appear and meal containers should make room for the scroll bar	As expected. However, there was a slow loading time	Pass
Search screen	Dragging scroll bar	MealThumb's should all move the correct way	As expected. Note: I did see some overlapping text from one of the MealThumb's	Pass

Search screen	Dragging scroll bar to very end	MealThumb's should stop moving, scroll bar should stop moving	Worked fine when the MealThumb's hit the bottom of the screen. However, when moving to the top it went past the stop line and look jolty	Fail
Search screen	Click back button	Navigate to the home screen	As expected	Pass
Search screen	Click any MealThumb	Navigate to the recipe screen	As expected	Pass
Recipe screen	Content height is more than the container height	Scroll bar should show and all contents on screen should have a smaller width to make space for it	As expected	Pass
Recipe screen	Click and drag scroll bar	Scroll bar should move contents on the screen	Contents weren't fitted to the scroll bar ratio. I was able to scroll past the top of the screen.	Fail
Recipe screen	Click back button	Navigate to the search screen	As expected.	Pass
Recipe screen	Upon entering	All contents should show as expected	Full recipe did not show	Fail

Running these tests helped me notice some of the flaws and discrepancies within my application. I was able to identify errors and bugs within the program so that I could resolve the issues that I found. I was able to fix some of the issues, but unfortunately I wasn't able to fix all of them.

Link to the testing video: https://drive.google.com/file/d/1n_vafeHTVGExU4C06UI3yZ-Oowf24Pgj/view?usp=share_link

Critical Reflection

Overall I am very happy with the results of my application, I learned a lot from the process and my knowledge of coding has improved. I think the application demonstrates a good knowledge of programming and some advanced programming skills. However, having said that I still think there are many ways in which it could have been improved. I was unsure about some of the program structure and I think that some of the contents of classes, especially in the screen classes could have been separated off further. I also think that some of the contents of the component classes potentially didn't need to be in their own class. I would have liked to have had another screen where the user could save their favourite meals, but I unfortunately ran out of time to do this.

The Gantt chart was a helpful part of the process. It helped me see the dependencies between different tasks and identify any potential roadblocks that could have affected the overall timeline. I was able to break down my project into smaller, manageable tasks and set clear deadlines for each one.

The majority of the tests passed as expected. However, some errors and bugs were shown during this process. I noticed that some of the responses from the Api were different to others, where some data was incomplete, some of different length strings and some data was completely missing. I was able to fix some of my coding errors and I would have liked to have had more time to thoroughly look through the database results and attempt to improve.

This has been a big milestone for me with programming and I am happy with the application I have created.